Artificial Intelligence

Lecture 1 - AI Background

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Course Objectives

1. Provide a concrete grasp of the fundamentals of various techniques and branches that currently constitute the field of Artificial Intelligence, e.g.,
    1. Agents
    2. Search
    3. Knowledge Representation
    4. Autonomous planning
    5. Reasoning under uncertainty

Outline

- Course overview
- What is AI?
- A brief history of AI
- The state of the art of AI

Course overview

- Introduction and Agents (Chapters 1,2)
- Search (Chapters 3,4,5,6)
- Logic (Chapters 7,8,9)
- Planning (Chapters 11,12)
- Reasoning under uncertainty ()

Human Intelligence:
How we think ie can perceive, understand, predict and manipulate world much more complicated than itself.
But AI goes further to build Intelligent entities.
AI has two aspects: Thinking and behavior

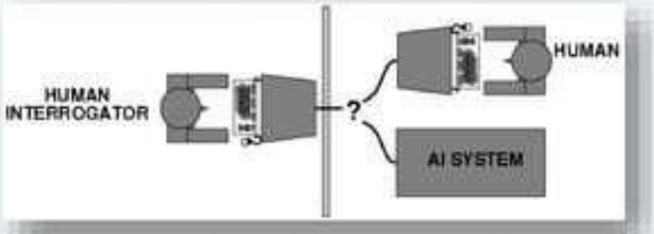
What is AI?

- Views of AI fall into four categories:
  - Systems that act like humans
  - Systems that think like humans
  - Systems that act rationally
  - Systems that think rationally

- In this course, we are going to focus on systems that act rationally, i.e., the creation, design and implementation of rational agents.

Acting Humanly: Turing Test

- Turing (1950) "Computing machinery and intelligence".
- A computer passes the test if a human interrogator, after posing some written questions, cannot tell whether the written responses come from a person or from a computer.



- Anticipated all major arguments against AI in following 50 years
- Little effort by AI researchers to pass the Turing Test

Turing Test

- **Major Components of Turing Test:**
  - Natural Language Processing: To enable it to communicate successfully in English.
  - Knowledge Representation: To store what it knows or hears.
  - Automated Reasoning: To use the stored information to answer questions and to draw conclusions.
  - Machine Learning: To adapt to new circumstances and to detect and extrapolate patterns.
- **Total Turing Test also includes:**
  - Computer Vision: To perceive objects
  - Robotics: To manipulate objects and move about

Thinking Humanly: Cognition

- Expressing the Theory of Mind as a Computer Program
  - GPS (Newell & Simon 1961) does not only need to solve the problems but should also follow human thought process

- Requires scientific theories of internal activities of the brain.
  - Cognitive Science: Predicting and testing behavior of human subjects
  - Cognitive Neuroscience: Direct identification from neurological data

Thinking Rationally: “Laws of Thought"

- Aristotle: First to codify “right thinking”
- Several Greek schools developed various forms of logic:
  - Notation and rules of derivation for thoughts
- By 1965, programs existed that could, in principle, solve any solvable problem described in logical notation.

- Problems:
  - Not easy to state informal knowledge in logical notation
  - Big difference between solving a problem "in principle" and solving it “in practice”
    - Problems with just a few hundred facts can exhaust the computational resources of any computer

Acting rationally: rational agent

- Rational behavior: doing the right thing
- The right thing: the optimal (best) thing that is expected to maximize the chances of achieving a set of goals, in a given situation
- Making correct inferences is sometimes part of being a rational agent
- Advantages over other approaches
  - More general than the "laws of thought" approach
  - More amenable to scientific development than are approaches based on human behavior or human thought
  - Standard of rationality is mathematically well defined and completely general

Rational Agents

- An agent is an entity that perceives and acts
- This course is about designing rational/intelligent agents
- Abstractly, an agent is a function from percept histories to actions:
  - f : P* -> A
- For any given class of environments and tasks, we seek the agent (or class of agents) with the optimal (best) performance
- Caveat: computational limitations make perfect rationality unachievable
  - So we attempt to design the best (most intelligent) program, under the given resources.

AI Prehistory

- Philosophy: Logic, methods of reasoning, mind as physical system, foundations of learning, language, rationality
- Mathematics: Formal representation and proof, Algorithms, Computation, (un)decidability, (in)tractability, probability
- Psychology: Adaptation, phenomena of perception and motor control, experimental techniques (with animals, etc.)
- Economics: Formal theory of rational decisions
- Linguistics: Knowledge representation, grammar
- Neuroscience: Plastic physical substrate for mental activity
- Control theory: Homeostatic systems, Stability, Simple optimal agent designs.

Abridged history of AI

- 1943 McCulloch & Pitts: Boolean circuit model of brain
- 1950 Turing's "Computing Machinery and Intelligence"
- 1956 Dartmouth: "Artificial Intelligence" adopted
- 1952-69 Look, Ma, no hands!
- 1950s Early AI programs, including Samuel's checkers program, Newell & Simon's Logic Theorist,
- 1965 Robinson's algo for logical reasoning
- 1966-73 AI discovers computational complexity Neural network research almost disappears
- 1969-79 Early development of knowledge-based systems
- 1980-- AI becomes an industry
- 1986-- Neural networks return to popularity
- 1987-- AI becomes a science
- 1995-- The emergence of intelligent agents.

McCulloch & Pitts (1943)

- Proposed a model of artificial neurons
- Each neuron is characterized as being "on" or"off,"
- Switch to "on" occurring in response to stimulation by a sufficient number of neighboring neurons.
- The state of a neuron was conceived of as "factually equivalent to a proposition
- Any computable function could be computed by some network of connected neurons
- All the logical connectives (and, or, not, etc.) could be implemented by simple net structures.
- McCulloch and Pitts also suggested that suitably defined networks could learn.
- First Neural Network Computer (1950)

Dartmouth (1956)

- 2 Month, 10 Man Study of AI
- Newell and Simon came up with a reasoning program, the Logic Theorist (LT)
- The program was able to prove most of the theorems in Chap 2, Principia Mathematica

Early Enthusiasm (1952 - 1969)

- GPS (thinking humanly)
- Herbert Gelemter (1959) constructed the Geometry Theorem Prover
- Arthur Samuel (1956) wrote a series of programs for checkers (draughts) that eventually learned to play at a strong amateur level
- LISP (1958) by John McCarthy

Dose of Reality (1966 - 1973)

- In almost all cases, these early systems turned out to fail miserably when tried out on wider selections of problems and on more difficult problems.
  - Intractability of problems

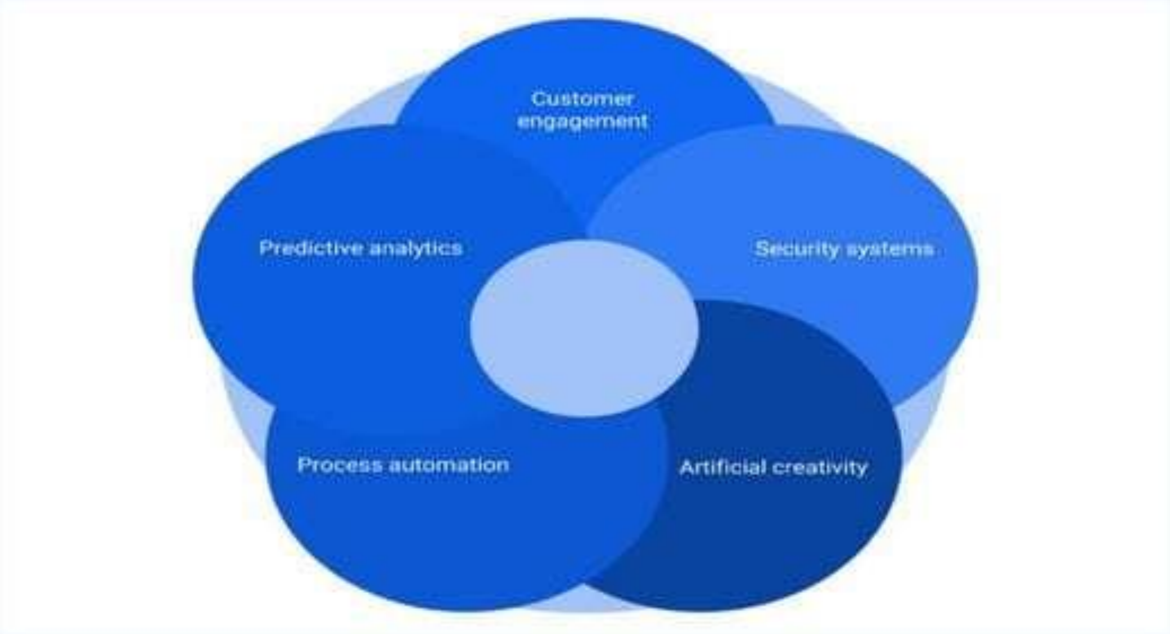
- Failure to come to grips with the "combinatorial explosion" was one of the main criticisms of AI contained in the Lighthill report (Lighthill, 1973), which formed the basis for the decision by the British goverrunent to end support for AI research

Knowledge Based Systems (1969 - 1979)

- DENDRAL
- MYCIN

Cutting Edge Application of AI

Applications of AI in different business areas



Construction

- AI-based applications have been widely being used in the construction sectors.
- The AI-based application will make the engineers more productive and capable of delivering high-quality work in the stipulated time frame.

Agriculture

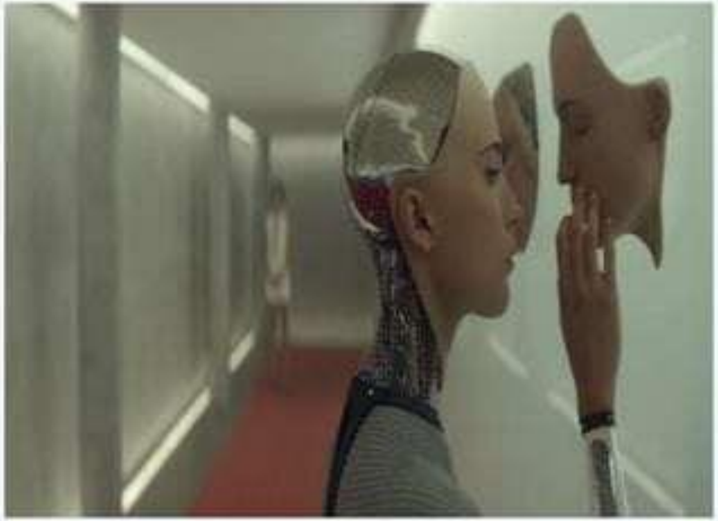
- Agriculture is one of the core sectors and we have been modifying the cultivation process to yield more from it.
- There are incredible opportunities for AI or machine learning in agriculture.
- The technologies like AI & IoT will be very useful in understanding a timely planting, getting predictions, using fertilizers, harvesting and the climate.

Sports

- The implementation of artificial intelligence in the sports industry is a game changer.
- There has been a huge demand for AI-based applications in the sports industry as it possesses significant capabilities.
- The implementation of the technology is certainly going to solve many major changes in the sports world and it could bring the true competition among the players and athletes.

Entertainment

- AI or machine learning has brought a big change in the entertainment industry.
- AI is everywhere and it is making a big difference in our lives.
- When it comes to the entertainment, the algorithms being used by various application make our life much simpler.
- AI has really changed the entertainment industry and it will make it more lively in the days to come.

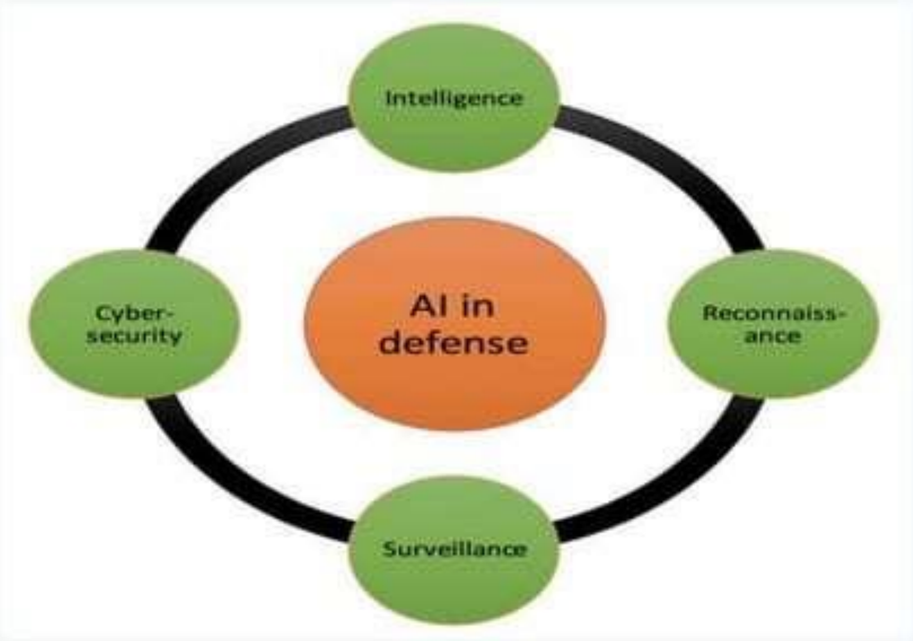


Life On Other Planets

- NASA is already using AI to look for life on other planets, which will be the key for "Mars 2020," the mission where the Red Planet will be explored more thoroughly.
- The devices they'll send, better known as rovers, will be able to explore Mars' terrain in more detail and reveal the properties of the planet's elements to determine the possibility of life with more certainty

Ways Global Defense Forces Use AI

- ❑ Military drones for surveillance
- ❑ Robot soldiers for combat
- ❑ Intelligent systems for awareness
- ❑ Secure web-portals for cybersecurity


Intelligence
Cyber-
security
AI in
defense
Reconnaiss-
ance
Surveillance

State of the art

- Deep Blue defeated the reigning world chess champion Garry Kasparov in 1997
- No hands across America (driving autonomously 98% of the time from Pittsburgh to San Diego)
- During the 1991 Gulf War, US forces deployed an AI logistics planning and scheduling program that involved up to 50,000 vehicles, cargo, and people
- NASA's on-board autonomous planning program controlled the scheduling of operations for a spacecraft
- Proverb solves crossword puzzles better than most humans.

Natural Language Processing

- Speech technologies
  - Automatic speech recognition (ASR)
  - Text-to-speech synthesis (TTS)
  - Dialog systems
- Language Processing Technologies
  - Machine Translation
  - Information Extraction
  - Informtation Retrieval
  - Text classification, Spam filtering.

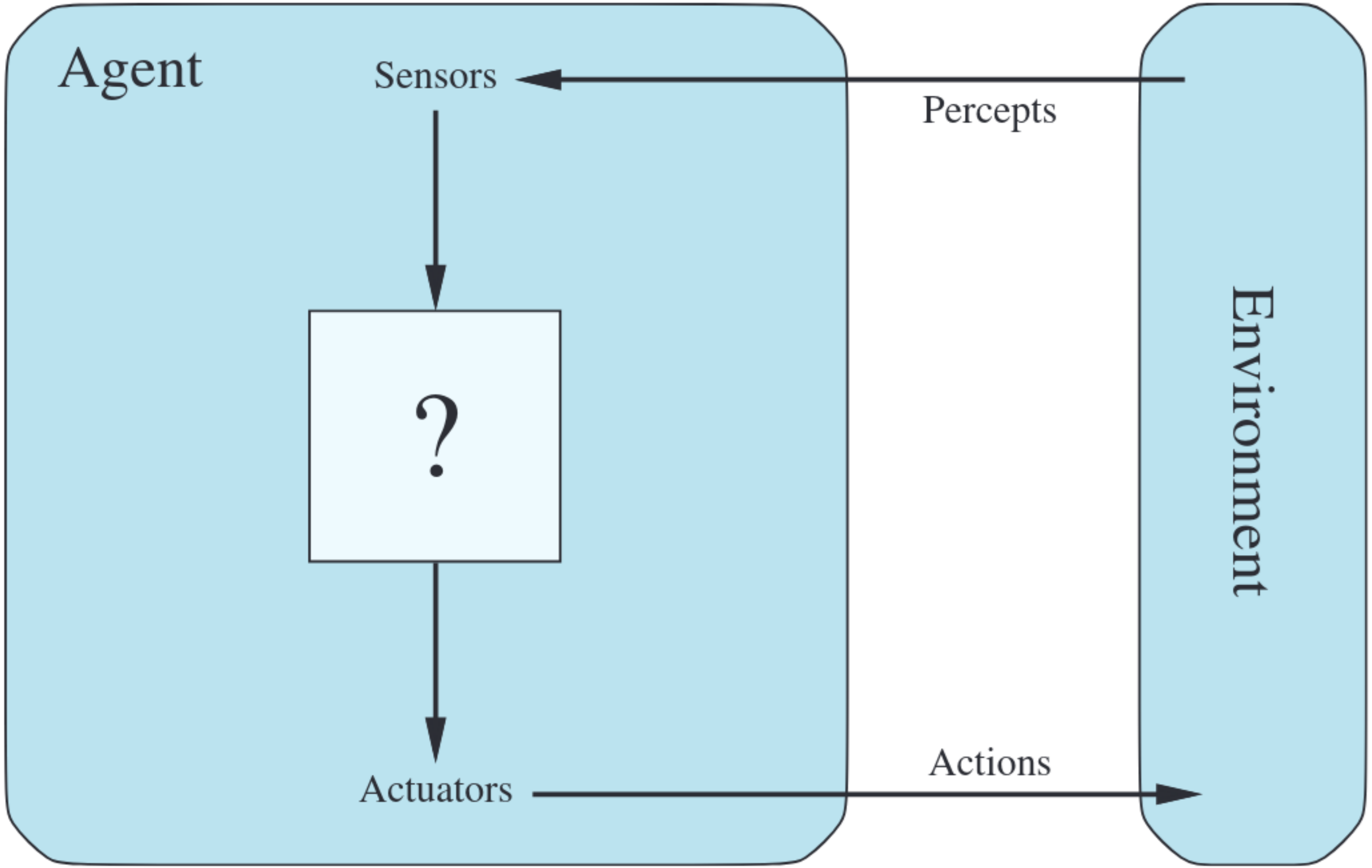
Robotics

Others..

- Computer Vision:
  - Object and Character Recognition
  - Image Classification
  - Scenario Reconstruction etc.
- Game-Playing
  - Strategy/FPS games, Deep Blue etc.
- Logic-based programs
  - Proving theorems
  - Reasoning etc.

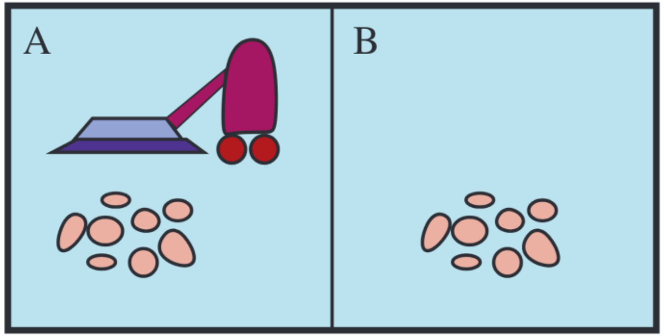
Intelligent Agents

- Agent
- Environment
- Sensors
- Actuators



Example - Vacuum Agent

Percept sequence	Action
$[A, Clean]$	$Right$
$[A, Dirty]$	$Suck$
$[B, Clean]$	$Left$
$[B, Dirty]$	$Suck$
$[A, Clean], [A, Clean]$	$Right$
$[A, Clean], [A, Dirty]$	$Suck$
$\vdots$	$\vdots$
$[A, Clean], [A, Clean], [A, Clean]$	$Right$
$[A, Clean], [A, Clean], [A, Dirty]$	$Suck$
$\vdots$	$\vdots$



Intelligent Agents

- Considerations...
    - How should it act?
    - How does it know it is doing what it is supposed to do?
    - How does it know not to make a mistake?
    - Should we monitor it at all times?
    - What happens if it is put in a different environment?

Rational Agents

- **Performance Measure** reports how “desirable” was an Agent’s action.
- Rationality depends on:
    - The performance measure that defines the criterion of success.
    - The agent’s prior knowledge of the environment.
    - The actions that the agent can perform.
    - The agent’s percept sequence to date.

PEAS - Task Environment

- PEAS (**P**erformance, **E**nvironment, **A**ctuators, **S**ensors) description of an environment:
    - Agent Type
    - Performance Measure
    - Environment
    - Actuators
    - Sensors

PEAS

Agent Type	Performance Measure	Environment	Actuators	Sensors
Taxi driver	Safe, fast, legal, comfortable trip, maximize profits, minimize impact on other road users	Roads, other traffic, police, pedestrians, customers, weather	Steering, accelerator, brake, signal, horn, display, speech	Cameras, radar, speedometer, GPS, engine sensors, accelerometer, microphones, touchscreen

Agent Type	Performance Measure	Environment	Actuators	Sensors
Medical diagnosis system	Healthy patient, reduced costs	Patient, hospital, staff	Display of questions, tests, diagnoses, treatments	Touchscreen/voice entry of symptoms and findings
Satellite image analysis system	Correct categorization of objects, terrain	Orbiting satellite, downlink, weather	Display of scene categorization	High-resolution digital camera
Part-picking robot	Percentage of parts in correct bins	Conveyor belt with parts; bins	Jointed arm and hand	Camera, tactile and joint angle sensors
Refinery controller	Purity, yield, safety	Refinery, raw materials, operators	Valves, pumps, heaters, stirrers, displays	Temperature, pressure, flow, chemical sensors
Interactive English tutor	Student's score on test	Set of students, testing agency	Display of exercises, feedback, speech	Keyboard entry, voice

Environment Properties

- Fully v/s Partially observable
- Single-agent vs. Multiagent
- Deterministic vs. Nondeterministic
- Episodic vs. Sequential
- Static vs. Dynamic
- Discrete vs. Continuous

Environment Examples

Task Environment	Observable	Agents	Deterministic	Episodic	Static	Discrete
Crossword puzzle	Fully	Single	Deterministic	Sequential	Static	Discrete
Chess with a clock	Fully	Multi	Deterministic	Sequential	Semi	Discrete
Poker	Partially	Multi	Stochastic	Sequential	Static	Discrete
Backgammon	Fully	Multi	Stochastic	Sequential	Static	Discrete
Taxi driving	Partially	Multi	Stochastic	Sequential	Dynamic	Continuous
Medical diagnosis	Partially	Single	Stochastic	Sequential	Dynamic	Continuous
Image analysis	Fully	Single	Deterministic	Episodic	Semi	Continuous
Part-picking robot	Partially	Single	Stochastic	Episodic	Dynamic	Continuous
Refinery controller	Partially	Single	Stochastic	Sequential	Dynamic	Continuous
English tutor	Partially	Multi	Stochastic	Sequential	Dynamic	Discrete

How much Intelligence?

- Agent v/s Programmer
    - Agent: How much does the agent have to do with determining its next action?
    - Programmer: How much predetermined actions are programmed?

Simple Reflex Agent

- Based only on the input of the sensor
- Rules are triggered (IF-ELSE rules)
- Predetermined actions mostly
- Random Behaviour sometimes

Model Based Agent

- Partially observable problems
- Based on the inputs of the sensor and knowledge of the world
- State of the world updates based on inputs
- Rules are triggered
- Intelligence is shared. (programmer can have big influence)
- Some non-deterministic behaviour

Goal Based Agent

- Goals are evaluated by the system (according to state and sensor data)
- Based on the inputs of the sensor and knowledge of the world
- Rules are triggered
- Intelligence is shared. (programmer has less influence)
- Mostly non-deterministic behaviour

Utility Based Agent

- Goals are evaluated according to **utility function**
- Based on the inputs of the sensor and knowledge of the world
- Rules are triggered
- Intelligence is shared. (programmer has less influence)
- Mostly non-deterministic behaviour

Learning Agents

- Can learn new information from environment
- Goal is to improve the performance
- Critic/Evaluation component
- Rewards or penalties
- Problem generator

AI Topics

- Search and Game Playing
- Probability of next action(HMM)
- Constraint Satisfaction
- Structured Representation

Thank You

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